Fiscal Year:	FY 2009	**Task Last Updated:**	FY 10/08/2009
PI Name:	Bloomberg, Jacob J. Ph.D.		
Project Title:	Development of Countermeasures to Enhance Sensorimotor Adaptation		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Task Description:	Astronauts develop adaptive responses to microgravity during space flight. Consequently these adaptive responses cause maladaptive disturbances in sensorimotor function when astronauts return to Earth and are re-exposed to gravity. During the acute phase of adaptation to novel gravitational environments on exploration class missions such sensorimotor disturbances will probably disrupt the ability to ambulate and perform functional tasks. These disturbances are likely to impair performance of complex tasks and could, therefore, affect astronaut safety and ultimate mission success. No operational countermeasure exists to mitigate the risks associated with space flight-induced sensorimotor disturbances. Therefore, this project will develop a comprehensive sensorimotor adaptability training program (SA training) to facilitate rapid adaptation to novel gravitational environments. This program will allow successful completion of operational tasks soon after landing on a planetary surface. We expect that SA training will facilitate restoration of functional capabilities when astronauts transition to different gravitational environments. Evidence in both the motor learning and clinical rehabilitation literature shows that training with task variability along with exposure to repeated sensorimotor challenges leads to faster adaptation that can be generalized to new environments. By applying these motor learning concepts for training astronauts we can enhance their ability to "learn how to learn" to adapt to new gravitational environments. Recent evidence suggests that SA training effects can be retained for at least one year. Given the in-flight logistic constraints on hardware volume and power as well as crew time, preflight training without or with a limited in-flight component would be preferable. To minimize cost and demands on crew time integrating SA training with existing exercise activities would also be preferable. Mission activities have special performance requirements unlike most terrestrial tasks. Therefore, a unique aspect of this study is that we will test the new countermeasures training approach with operationally oriented, functional tests that are directly related to crew performance needs. This project will develop a cost-effective, time-efficient countermeasure that will help overcome maladaptive sensorimotor disturbances enabling performance of functional tasks during transition to new gravitational environments after short and long transit durations during exploration class missions. The data obtained in this study will aid in the design of a countermeasures system used to support exploration-class space missions. The operational version of this countermeasure will use a virtual reality system coupled with a treadmill or stepper system on a passive or active motion base that will allow the user to walk and also perform complex manual tasks in a varied and interesting virtual environment. This training system can serve as a pre-and inflight training tool providing sufficient sensorimotor challenge to crewmembers to maximize their motor response adaptability, facilitating the adaptive transition to partial or unit gravity. During this reporting period we conducted a study to determine if sensorimotor adaptability (SA) training using visual flow and support surface motion produces improved performance in a novel sensory environment and demonstrate short (days) and long-term retention (up to 6 month) of SA training. Twenty subjects (10 experimental, 10 control) completing three training sessions. Upon completion of training subjects performed the Transfer/Retention Test consisting of the subjects viewing a novel visual scene, as well as experiencing a novel support surface movement, not previously experienced during training. The Transfer/Retention Test was performed 20 minutes, 1 week 1 month, 3 months, and 6 months after the final training session. A total of 140 training and test sessions were completed. Preliminary data indicate that: 1) Subjects who received the SA training had improved locomotor performance and faster reaction times when exposed to the novel combination of incongruent visual flow and support surface motions. These data indicate that SA training enhanced adaptive responses to novel sensorimotor challenges. 2) Subjects who received the SA training retained the ability to adapt to incongruent sensory stimuli over a six-month period. 3) Subjects who experienced higher visual dependency for postural stability control during training experienced decreased postural stability and reaction times when exposed to a novel combination of incongruent visual flow and support surface motions. This suggests that individual sensory biases can be identified during training and individual training prescription developed to enhance sensorimotor adaptability. Proposed research plan for the upcoming year: The next experimental series will focus on examining how SA training can be used to increase adaptability over a wide range of functional tasks relevant to astronaut planetary EVAs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Sensorimotor adaptability training programs have earthbound application in rehabilitation of patients with balance disorders, and for fall prevention training among seniors. The most important factors underlying morbidity in the older adult population are injurious falls and the restriction of activity due to falls. Approximately 25-35% of community-dwelling persons older than 65 years fall at least once a year, and approximately 40-50% of fallers experience two or more falls. The outcome is that 40% of all nursing home admissions are due to injuries due to falls. Age-related sensory changes and deterioration in ability to compensate in older adults can contribute to increase incidence of falling. We previously conducted a study to determine if balance training using variation in visual flow during treadmill exercise improves functional mobility in healthy older adults who were experiencing age-related postural instabilities (Buccello-Stout et al. 2008). The results showed that subjects who were exposed to varied visual flow during treadmill walking significantly improved their ability to negotiate an obstacle course after training compared to another elder group who only walked on a treadmill for the same amount of time. Importantly, the training benefit was retained when subjects were tested four weeks later. This study confirms that adaptability training developed for use by astronauts can also be used to improve balance and gait performance in elder subjects and points to the general applicability of this type of training in different clinical populations.

Task Progress:	1) During this reporting period we conducted a study to determine if sensorimotor adaptability (SA) training using visual flow and support surface motion produces improved performance in a novel sensory environment and demonstrate short (days) and long-term retention (up to 6 month) of SA training. Twenty subjects (10 experimental, 10 control) completing three training sessions. Upon completion of training subjects performed the Transfer/Retention Test consisting of the subjects viewing a novel visual scene, as well as experiencing a novel support surface movement, not previously experienced during training. The Transfer/Retention Test was performed 20 minutes, 1 week 1 month, 3 months, and 6 months after the final training session. A total of 140 training and test sessions were completed. The entire experimental sequence took 7 months to complete. 2) In the next set of studies the efficacy of SA training will be determined using a set of operationally oriented functional tests performed during exposure to a novel set of visual and vestibular sensory distortions not previously experienced by the subjects during training. These sensory distortions will test the efficacy of SA training to facilitate rapid reorganization of behavioral responses to novel sensory environments. To produce a novel visual distortion we will use a head-mounted virtual reality (VR) system that will receive input from a head mounted camera. We will be able to manipulate the visual input received by the subject by tilting the camera relative to the head or adding a minifying or magnifying lens to the head mounted camera. This head mounted VR system is superior to using goggles with optics because it allows a wide variation of visual distortions and also permits a wider field of view. Vestibular distortion will be produced by Galvanic vestibular stimulation (GVS). GVS entails electrical stimulation to the vestibular labyrinth via surface electrodes placed over the mastoid bones that pass small currents that activate primary vestibular afferents. It has been used to stimulate the vestibular labyrinth artificially for laboratory studies of human spatial orientation, postural control and locomotion. We have previously shown that GVS produces behavioral changes in balance function, gaze and locomotor control that are similar to those observed in post-flight astronauts (Moore et al. 2006). During this reporting period we have developed the VR hardware necessary to provide the visual distortions described above. We have also designed and manufactured a portable GVS unit. This required significant interaction and multiple reviews by NASA safety officers during the design phase to ensure that the GVS unit met all of NASA's safety standards. This process is now complete and several GVS units are being built to support this study.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Brady RA, Brian BT, Bloomberg JJ. "Strategies of healthy adults walking on a laterally oscillating treadmill." Gait Posture. 2009 Jun;29(4):645-9. PMID: 19233653 ; http://dx.doi.org/10.1016/j.gaitpost.2009.01.010 , Jun-2009
Significant Media Coverage	Bloomberg JJ. "Participated in the production of a vodcast for the online educational program NASA 360 entitled, The Mind Body Connection." NASA 360 vodcast entitled, The Mind Body Connection, August 2009. (http://www.nasa.gov/multimedia/podcasting/nasa360/nasa360-archives.html). Also, http://www.nasa.gov/mp4/312373main_nasa360-0107-hi.mp4 ; accessed 7/20/15,, Aug-2009